Static Gesture Classification — Run Report

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Data paths

Training data root	None
External test root	/home/jestin/ThesisRepo/ML/NewTestData/Nat/Static
Report output dir	/home/jestin/ThesisRepo/ML/saved_reports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	176 channels
Classes	7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 5.0 Hz · order 4 · fs 30.0 Hz
Normalisation	standard
Feature mode	stats → feature dim 1408
Test size · seed · CV folds	0.2 · random_state=42 · 10-fold stratified
Sample counts	train (post-aug)=500 · test=63

Augmentation (training only)

Enabled	yes
Copies per train trial	1
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.8, 1.2)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (10-fold)	CV std	Hold-out test acc
Stacking (SVM+RF+KNN+LogReg)	1.0000	0.0000	0.9524
Voting Soft (SVM+RF+KNN+LogReg)	0.9980	0.0060	0.9365
SVM (Linear)	0.9980	0.0060	0.9206
Random Forest	0.9960	0.0120	0.9206
Logistic Regression	0.9980	0.0060	0.9206
SVM (RBF)	1.0000	0.0000	0.9048
KNN	0.9300	0.0371	0.7937
LDA	0.9460	0.0429	0.3492

Per-class hold-out F1

Class	Stacking	Voting (soft)	SVM (Linear)	Random Forest	Logistic Regression	SVM (RBF)	KNN	LDA	Support
Double_Closed_Fist	1.000	1.000	0.952	1.000	0.952	0.952	0.857	0.545	10
Double_Nothing	0.933	0.857	0.857	0.769	0.857	0.889	0.615	0.154	8
Double_Okay	0.947	0.947	0.900	1.000	0.900	0.947	0.556	0.364	9
Double_Open_Palm	0.941	0.941	0.941	1.000	0.941	0.875	0.778	0.267	9
Double_Phone	0.941	0.941	0.875	0.875	0.875	0.714	0.875	0.308	9
Double_Pistol	1.000	0.947	1.000	0.857	1.000	0.947	0.857	0.353	9
Double_Spiderman	0.900	0.900	0.900	0.900	0.900	0.947	0.947	0.333	9
macro avg	**0.952**	**0.933**	**0.918**	**0.914**	**0.918**	**0.896**	**0.784**	**0.332**	**63**

External test (NewTestData)

Algorithm	External test acc
SVM (RBF)	1.0000
SVM (Linear)	1.0000
Random Forest	1.0000
KNN	1.0000
Logistic Regression	1.0000
Voting Soft (SVM+RF+KNN+LogReg)	1.0000
Stacking (SVM+RF+KNN+LogReg)	1.0000
LDA	0.9189

Per-class external accuracy

Class	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	Voting (soft)	Stacking	LDA	Support
Double_Closed_Fist	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	6
Double_Nothing	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	5
Double_Okay	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.800	5
Double_Open_Palm	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	5
Double_Phone	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.800	5
Double_Pistol	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.833	6
Double_Spiderman	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	5